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MATHEMATICS TEST - CHAPTER 8: GEOMETRY OF STRAIGHT LINES

Name:		Roll No.		Section:

Total Time: 2:30 Hour **Total Marks: 150**
Time Allowed: 150 Minutes

PAPER OBJECTIVE
SECTION-A (MARKS 150)

NOTE: Section-A is compulsory. All parts of this section are to be answered on these pages and handed over to the center superintendent. Deleting / overwriting is not allowed. Do not use lead pencil.

Q.1 Fill the relevant bubble for each part. All parts carry one mark.

Questions	A	B	C	D	A B C D
1. The inclination of a line is the angle it makes with the	Positive y-axis	Positive x-axis (anticlockwise)	Negative x-axis	Origin	○●○○
2. The inclination of a horizontal line is	0°	90°	180°	45°	●○○○
3. The inclination of a vertical line is	0°	180°	45°	90°	○○○●
4. A line parallel to the x-axis is called a	Horizontal line	Vertical line	Oblique line	Normal line	●○○○
5. A line parallel to the y-axis is called a	Horizontal line	Oblique line	Vertical line	Tangent	○○●○
6. A line neither horizontal nor vertical is called	Parallel	Oblique	Perpendicular	Normal	○●○○
7. Gradient is also called	Intercept	Distance	Angle	Slope	○○○●
8. The gradient of a line is	tan of its inclination	sin of its inclination	cos of its inclination	cot of its inclination	●○○○
9. The slope of a line is denoted by	c	m	p	k	○●○○
10. The slope of the line through (x_1, y_1) and (x_2, y_2) is	$(y_2 - y_1)/(x_2 - x_1)$	$(x_2 - x_1)/(y_2 - y_1)$	$(y_2 + y_1)/(x_2 + x_1)$	$(y_2 - y_1)(x_2 - x_1)$	●○○○
11. The slope of a horizontal line is	1	Undefined	−1	0	○○○●
12. The slope of a vertical line is	0	Undefined	1	−1	○●○○
13. A line with positive slope makes an angle with the x-axis that is	Acute	Obtuse	Right	Straight	●○○○
14. A line with negative slope makes an angle with the x-axis that is	Acute	Zero	Obtuse	Right	○○●○

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Questions	A	B	C	D	A B C D
15. Parallel non-vertical lines have	Opposite slopes	Reciprocal slopes	Equal slopes	Zero slopes only	○○●○
16. For perpendicular lines, $m_1 \times m_2$ equals	1	−1	0	2	○●○○
17. If slope of a line is m, slope of a perpendicular line is	1/m	−1/m	−m	m	○●○○
18. Three points are collinear if the gradients of AB, BC and AC are	Equal	Zero	Opposite	Reciprocal	●○○○
19. The equation of a line parallel to the x-axis is	x = a	y = b	y = mx	x + y = 0	○●○○
20. The equation of a line parallel to the y-axis is	y = b	x = a	y = mx	x + y = 0	○●○○
21. The equation of the x-axis is	x = 0	x + y = 0	x = y	y = 0	○○○●
22. The equation of the y-axis is	y = 0	x + y = 0	x = 0	x = y	○○●○
23. The line y = b is above the x-axis if	b > 0	b < 0	b = 0	b ≠ 0	●○○○
24. The line y = b is below the x-axis if	b > 0	b = 0	b ≥ 0	b < 0	○○○●
25. The line x = a is to the right of the y-axis if	a < 0	a = 0	a > 0	a ≠ 0	○○●○
26. The line x = a is to the left of the y-axis if	a > 0	a = 0	a ≥ 0	a < 0	○○○●
27. If a line meets the x-axis at (a, 0), then a is called its	y-intercept	x-intercept	Slope	Gradient	○●○○
28. If a line meets the y-axis at (0, b), then b is called its	x-intercept	Slope	Gradient	y-intercept	○○○●
29. Slope of the line joining (0, 3) and (3, −4) is	7/3	3/7	−7/3	−4/3	○○●○
30. Equation of the vertical line through (3, 5) is	x = 3	x = 5	y = 3	x = −3	●○○○
31. Slope of the line joining (7, −8) and (−1, −4) is	−2	1/2	2	−1/2	○○○●

Questions	A	B	C	D	A B C D
32. If the slope of a line is 2, the slope of a line perpendicular to it is	1/2	−1/2	2	3/2	○●○○
33. Equation of the horizontal line through (2, 3) is	x = 3	y = 3	y = −3	x = 2	○●○○
34. Slope of the line joining (2, 6) and (5, 8) is	2	3/2	2/3	−2/3	○○●○
35. Slope of the line joining (−5, −6) and (6, −3) is	3/11	−11/3	14/11	11/3	●○○○
36. Pollution index: 80 ppm at 10:00 and 110 ppm at 14:00. Rate (ppm/hr) =	0.13	4	15	7.5	○○○●
37. Slope of the line joining (1, 7) and (−3, −1) is	3	−2	−1/2	2	○○○●
38. Slope of the line joining (−3, −1) and (4, 7) is	8/7	15/7	−8/7	−7/8	●○○○
39. Meter shows 44 units; 18 units/day. Units by 30 September (after 30 days) =	562	584	540	18	○●○○
40. The line 5x − 6 = 0 is parallel to the	y-axis	x-axis	line y = x	line x + y = 0	●○○○
41. The inclination of a line with slope −0.364 is	120°	90°	160°	60°	○○●○
42. The gradient of a line with inclination 45° is	1	0	−1	1/√3	●○○○
43. If the slope of a line is 6, the slope of a line perpendicular to it is	7/6	6	1/6	−1/6	○○○●
44. Slope of the line through A(4, 2), B(8, 9) is	11/4	4/7	7/4	−7/4	○○●○
45. A line has slope 3/4. A perpendicular line has slope	4/3	3/4	−4/3	5/3	○○●○
46. Slope of the line joining (7, −5) and (4, 1) is	1/2	−2	−1	2	○●○○
47. Slope of the line joining (−2, 0) and (−4, −6) is	−1/3	3	1/3	4	○●○○
48. Slope of the line joining (1, 2) and (4, −3) is	−2/3	−3/5	−5/3	3/5	○○●○

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Questions	A	B	C	D	A B C D
49. A line has slope 4/5. A parallel line has slope	−5/4	4/5	−4/5	5/5	○●○○
50. Pollution p = 7.5t + 5. The y-intercept (5) means pollution at	Noon is 5 ppm	12 am is 5 ppm	10 am is 5 ppm	Rate is 5 ppm/h	○●○○
51. Slope of the line joining (−1, 2) and (−8, 3) is	−1/7	7	−7	6/7	●○○○
52. A line has slope 2/3. A perpendicular line has slope	2/3	−3/2	4/2	−2/3	○●○○
53. The gradient of a line with inclination 120° is	−√3	−1	√3	1/√3	●○○○
54. The line 5x − ky − 3 = 0 passes through (1, 2). Then k =	1	5	−1	2	●○○○
55. Slope of the line joining (4, 8) and (−5, −3) is	−9/11	−11/9	9/11	11/9	○○○●
56. If the slope of a line is 8, the slope of a line perpendicular to it is	−8	8	1/8	−1/8	○○○●
57. Slope of the line joining (6, −1) and (3, 4) is	−2/3	3/5	−5/3	−3/5	○○●○
58. Slope of the line joining (−8, 2) and (3, 7) is	11/5	5/11	16/11	−5/11	○●○○
59. The y-intercept of the line x + y = 5 is	0	1	5	−5	○○●○
60. Slope of a line is 5. Slope of a perpendicular line is	1/5	5	−1/4	−1/5	○○○●
61. Slope of the line joining (3, 6) and (1, −4) is	1/5	6	5	−5	○○●○
62. For A(−2, 6), B(7, −3), the slope of a line perpendicular to AB is	1	−9	−1	9	●○○○
63. Slope of the line joining (−6, −6) and (−2, 1) is	−4/7	4/7	7/4	−7/4	○○●○
64. Equation of the horizontal line through (2, −2) is	x = −2	y = 2	x = 2	y = −2	○○○●
65. Pollution p = 7.5t + 5. At 12 pm (t = 12), p =	95	85	90	105	●○○○

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Questions	A	B	C	D	A B C D
66. The slope-intercept form of a line is	$y - y_1 = m(x - x_1)$	$x/a + y/b = 1$	$ax + by + c = 0$	$y = mx + c$	○○○●
67. The point-slope form of a line is	$y - y_1 = m(x - x_1)$	$y = mx + c$	$x/a + y/b = 1$	$x \cos \alpha + y \sin \alpha = p$	●○○○
68. The two-intercept form of a line is	$y = mx + c$	$x/a + y/b = 1$	$y - y_1 = m(x - x_1)$	$x \cos \alpha + y \sin \alpha = p$	○●○○
69. The normal form of a line is	$x \cos \alpha + y \sin \alpha = p$	$x/a + y/b = 1$	$y = mx + c$	$y - y_1 = m(x - x_1)$	●○○○
70. The symmetric form of a line is	$(x - x_1)/\cos \alpha = (y - y_1)/\sin \alpha$	$y = mx + c$	$x/a + y/b = 1$	$ax + by + c = 0$	●○○○
71. The general form of a linear equation in two variables is	$ax^2 + by + c = 0$	$ax + b = 0$	$ax = by^2$	$ax + by + c = 0$	○○○●
72. In $ax + by + c = 0$, the constants a and b	Must be equal	Must be positive	Must be integers	Cannot both be zero	○○○●
73. A linear equation in two variables represents a	Circle	Straight line	Parabola	Point	○●○○
74. The slope of $ax + by + c = 0$ ($b \neq 0$) is	a/b	$-b/a$	b/a	$-a/b$	○○○●
75. The y-intercept of $ax + by + c = 0$ ($b \neq 0$) is	$-c/b$	c/b	$-c/a$	c/a	●○○○
76. The x-intercept of $ax + by + c = 0$ ($a \neq 0$) is	c/a	$-c/a$	$-c/b$	c/b	○●○○
77. Slope of $ax + by + c = 0$ equals	(coeff of x)/(coeff of y)	−(coeff of y)/(coeff of x)	−(coeff of x)/(coeff of y)	c	○○●○
78. The line $y = mx$ passes through the	Point (1, 0)	Point (0, 1)	Point (1, 1) always	Origin	○○○●
79. Slope-intercept form of a line with $m = 5$ and $c = 3$ is	$y = 3x + 5$	$y = 5x + 3$	$y = 5x - 3$	$x = 5y + 3$	○●○○
80. Slope of the line $3x + 2y + 2 = 0$ is	2/3	−2/3	−1/2	−3/2	○○○●
81. $x \cos 150° + y \sin 150° = 10$: slope of the line is	$\sqrt{3}$	$-\sqrt{3}$	$1/\sqrt{3}$	$-1/\sqrt{3}$	●○○○
82. Intersection of $4x + 3y - 22 = 0$ and $x - y + 5 = 0$ is	(1, −6)	(6, 1)	(1, 6)	(2, 6)	○○●○

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Questions	A	B	C	D	A B C D
83. Point of intersection of $9x - 7y = 0$ and $8x - 11y = 0$ is	(0, 7)	(0, 0)	(9, 0)	(7, 9)	○●○○
84. Intersection of $x - y - 9 = 0$ and $3x + 2y - 2 = 0$ is	(4, −4)	(4, 5)	(−5, 4)	(4, −5)	○○○●
85. Line through (1, 0) and (−4, −5) has equation	$x - y = 0$	$x + y + 1 = 0$	$x - y - 1 = 0$	$x - y + 1 = 0$	○○●○
86. Line through (−5, 1) perpendicular to line through (0, 9) and (−2, 6) is	$2x - 3y + 7 = 0$	$3x + 2y + 7 = 0$	$2x + 3y + 7 = 0$	$2x + 3y - 7 = 0$	○○●○
87. x-intercept of the line $5x - y - 4 = 0$ is	9/5	−4/5	5	4/5	○○○●
88. A point on the line $y = 2/3x - 3$ is	(−2, −10/3)	(−1, −13/3)	(−2, 13/3)	(−2, −13/3)	○○○●
89. Intersection of $x + 3y + 10 = 0$ and $2x - y - 1 = 0$ is	(1, 3)	(−1, −3)	(0, −3)	(−1, −2)	○●○○
90. Line with slope −3 and y-intercept 1 has equation	$y = 3x + 1$	$y = -1/3x + 1$	$y = -3x - 1$	$y = -3x + 1$	○○○●
91. Line through (3, −2) perpendicular to a line of slope 3 has equation	$x + 3y + 3 = 0$	$x + 3y - 3 = 0$	$3x + y - 7 = 0$	$x + 3y + 9 = 0$	●○○○
92. A point on the line $y = x - 5$ is	(1, −4)	(−1, −4)	(2, −4)	(1, 4)	●○○○
93. Intersection of $2x - y + 3 = 0$ and $x + y = 0$ is	(−1, 2)	(1, −1)	(−1, 1)	(0, 1)	○○●○
94. Line with slope 3/2 and y-intercept −7 has equation	$y = -3/2x - 7$	$y = 3/2x + 7$	$y = 3/2x - 7$	$y = 3/2x - 6$	○○●○
95. Point of intersection of $x + 2y = 3$ and $3x - y = 2$ is	(1, 1)	(3, 2)	(−1, 1)	(1, −1)	●○○○
96. Line with slope 3 and y-intercept 4 has equation	$y = 4x + 3$	$y = 3x + 4$	$y = -3x + 4$	$y = 1/3x + 4$	○●○○
97. Line with x-intercept 3 and y-intercept 9 has equation	$x + 3y - 9 = 0$	$3x + y - 9 = 0$	$3x + y + 9 = 0$	$3x + 9y - 1 = 0$	○●○○
98. Line with slope −3 and y-intercept −1 has equation	$y = -3x + 1$	$y = -3x$	$y = -3x - 1$	$y = -x - 3$	○○●○
99. Intersection of $x - y + 4 = 0$ and $2x - 3y + 11 = 0$ is	(3, −1)	(1, −3)	(−1, −3)	(−1, 3)	○○○●

Questions	A	B	C	D	A B C D
100. Intersection of $4x - 3y - 8 = 0$ and $x - y - 2 = 0$ is	(0, 2)	(2, 1)	(2, 0)	(3, 0)	○○●○
101. Slope of the line $2x + y + 2 = 0$ is	−2	2	−1/2	−1	●○○○
102. Line with y-intercept −7 and parallel to a line of slope 4 is	$4x + y - 7 = 0$	$x - 4y - 7 = 0$	$4x - y - 7 = 0$	$4x - y + 7 = 0$	○○●○
103. The slope of $4x - 3y + 9 = 0$ is	−4/3	3	4/3	3/4	○○●○
104. Line with slope −3/2 and y-intercept −7 has equation	$y = -3/2x + 7$	$y = 3/2x - 7$	$y = -7x - 3/2$	$y = -3/2x - 7$	○○○●
105. Line through (3, −2), perpendicular to a line of slope 3 is	$x - 3y + 3 = 0$	$x + 3y + 3 = 0$	$3x + y + 3 = 0$	$x + 3y - 3 = 0$	○●○○
106. Intersection of $2x - y + 2 = 0$ and $x - y - 1 = 0$ is	(3, 4)	(−3, −3)	(3, −4)	(−3, −4)	○○○●
107. The line $y = 5x - 3$ is written in	Slope-intercept form	Point-slope form	Two-intercept form	Two-point form	●○○○
108. Line through (−6, 3) and (0, −3) has equation	$x - y + 3 = 0$	$x + y + 3 = 0$	$x + y + 2 = 0$	$x + y - 3 = 0$	○●○○
109. Slope of the line $2x - 3y = 0$ is	0	−3/2	2/3	3/2	○○●○
110. Intersection of $4x - 3y + 26 = 0$ and $3x + y = 0$ is	(2, −6)	(−2, −6)	(−2, 6)	(6, −2)	○○●○
111. Line through (0, 4) parallel to a line of slope 2/3 has equation	$2x + 3y - 12 = 0$	$3x - 2y + 8 = 0$	$2x - 3y - 12 = 0$	$2x - 3y + 12 = 0$	○○○●
112. Line with slope −3 and y-intercept −4 has equation	$y = -3x + 4$	$y = 3x - 4$	$y = -3x - 3$	$y = -3x - 4$	○○○●
113. A point on the line $y = 2x + 2$ is	(−1, 1)	(1, 0)	(−1, 0)	(0, 0)	○○●○
114. Line through (−1, −3) with slope 5 has equation	$x - 5y + 2 = 0$	$5x - y - 8 = 0$	$5x + y - 2 = 0$	$5x - y + 2 = 0$	○○○●
115. y-intercept of the line $5x - y + 4 = 0$ is	−20	−4	4	8	○○●○
116. Line through (5, 0) and (−4, 4) has equation	$4x + 9y - 20 = 0$	$4x + 9y - 29 = 0$	$81x + 36y - 80 = 0$	$4x - 9y - 20 = 0$	●○○○

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Questions	A	B	C	D	A B C D
117. Line with x-intercept 3 and y-intercept 4 has equation	$4x - 3y - 12 = 0$	$4x + 3y - 12 = 0$	$3x + 4y - 1 = 0$	$4x + 3y + 12 = 0$	○●○○
118. Slope of the line $3x + y + 2 = 0$ is	-3	$1/3$	$-1/3$	-2	●○○○
119. Line with slope 2 and y-intercept −6 has equation	$y = -6x + 2$	$y = 2x + 6$	$y = 2x - 6$	$y = -2x - 6$	○○●○
120. Which line has slope 2/3?	$2x - 3y = 2$	$2x + 3y = 0$	$3x - 2y = 1$	$3x + 2y = 3$	●○○○
121. Line through (4, 3) and (6, −3) has equation	$3x + y - 16 = 0$	$3x + y - 15 = 0$	$x + 3y - 45 = 0$	$3x - y + 15 = 0$	○●○○
122. Intersection of $4x + 3y - 2 = 0$ and $3x - 2y - 10 = 0$ is	$(2, 2)$	$(-2, 2)$	$(2, -2)$	$(-2, -2)$	○○●○
123. Line through (4, −2) and (−6, 4) has equation	$3x + 5y - 2 = 0$	$3x - 5y + 2 = 0$	$3x - 5y - 2 = 0$	$3x + 5y + 2 = 0$	●○○○
124. Intersection of $4x + 3y - 21 = 0$ and $2x + 3y - 15 = 0$ is	$(3, 3)$	$(-3, -3)$	$(-3, 3)$	$(3, -3)$	●○○○
125. Intersection of $2x + y + 5 = 0$ and $x - 3y + 6 = 0$ is	$(-3, -1)$	$(3, -1)$	$(-3, 1)$	$(1, -3)$	○○●○
126. Line with x-intercept −2 and y-intercept −6 has equation	$3x + y + 6 = 0$	$x + 3y + 6 = 0$	$3x + y - 6 = 0$	$2x + 6y + 1 = 0$	●○○○
127. Line with x-intercept 6 and y-intercept −4 has equation	$6x - 4y - 1 = 0$	$2x - 3y - 12 = 0$	$2x - 3y + 12 = 0$	$2x + 3y - 12 = 0$	○●○○
128. The angle θ from l_1 to l_2 is given by $\tan\theta =$	$(m_1 - m_2)/(1 + m_1m_2)$	$(m_2 + m_1)/(1 - m_1m_2)$	$(m_2 - m_1)/(1 + m_1m_2)$	$m_2 - m_1$	○○●○
129. If $l_1 \parallel l_2$, then $\tan\theta$ is	1	0	∞	−1	○●○○
130. If $l_1 \perp l_2$, then $1 + m_1m_2$ is	1	0	−1	2	○●○○
131. Angles between lines are measured from the	Negative x-axis	Positive y-axis, clockwise	Origin, clockwise	Positive x-axis, anticlockwise	○○○●
132. Two non-parallel lines intersect at	Exactly one point	No point	Two points	Infinitely many points	●○○○
133. Solution of two parallel lines	Is one point	Does not exist	Is two points	Is infinite	○●○○

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Questions	A	B	C	D	A B C D
134. Infinitely many lines can pass through	Two points	Three points	A point	Zero points	○○●○
135. Slopes of two lines are 0 and ∞. The angle between them is	0°	45°	180°	90°	○○○●
136. Median of a triangle joins a vertex to the	Foot of the opposite altitude	Another vertex	Midpoint of the opposite side	Centroid only	○○●○
137. The sum of interior angles of a triangle is	360°	90°	270°	180°	○○○●
138. A right triangle has one angle of	60°	45°	120°	90°	○○○●
139. The line $5x - 6 = 0$ is	Parallel to the y-axis	Parallel to the x-axis	Through the origin	Equally inclined to axes	●○○○
140. Line $x - 2y = 2$: its y-intercept is	-1	1	2	-2	●○○○
141. Angle from l_1 to l_2 with slopes 1/2 and 3 is	135°	45°	22.5°	60°	○●○○
142. Median of triangle with vertices (0, 4), (4, 6), (−2, −3) from (0, 4) goes to the midpoint	(2, 5)	(−1, 1/2)	(1, 3)	(1, 3/2)	○○○●
143. Perpendicular bisector of the segment joining (0, 6) and (2, −2) passes through	(0, 6)	(1, 2)	(2, −2)	(1, 4)	○●○○
144. A(0, 4), B(4, 6): midpoint of AB is	(4, 10)	(1, 3/2)	(2, 10)	(2, 5)	○○○●
145. For lines $x + 2y = 0$ and $x - y = 3$, the point of intersection is	(1, 1)	(−2, 1)	(2, −1)	(3, 0)	○○●○
146. Line $x - 2y = 2$: its slope is	$-1/2$	1/2	2	-2	○●○○
147. For triangle A(−2, 0), B(3, 0), C(6, 5), the interior angle at B is about	32°	121°	27°	59°	○●○○
148. Slopes 0 and 1. The angle from l_1 to l_2 is	0°	90°	45°	135°	○○●○
149. Angle from l_1 to l_2 with slopes -1 and 2 is	108.43°	71.57°	18.43°	54.22°	●○○○
150. The perpendicular bisector of the segment joining (0, 6) and (2, −2) has slope	1/4	−4	4	−1/4	●○○○

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